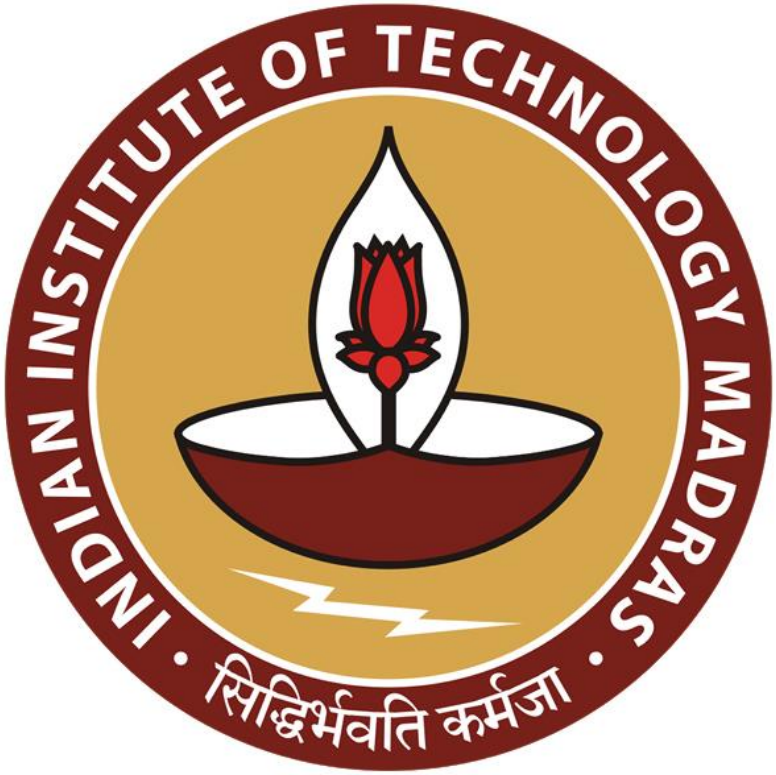
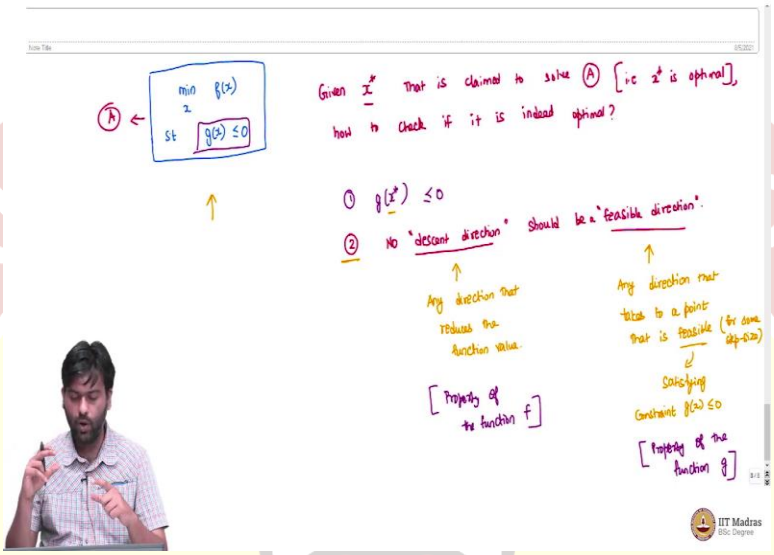


IIT Madras

ONLINE DEGREE

Constrained Optimization (Part – 1)
Assistant. Professor Arun Rajkumar
Department of Computer Sciences and Engineering
Indian Institute of Technology, Madras

(Refer Slide Time: 0:15)



Welcome back, what we want to do now is solve a constrained optimization problem with one single inequality constraint. Let me write the term for you, so we want to minimize x f(x) such that g(x) is less than or equal to 0. Of course, we will see when you actually do a machine learning course or you will see a lot of problems being posed as constraint optimization problems, so it is important for us to understand what are the optimization based ideas that can be used to solve constrained optimization problems.

We know that if there are no constraints, if x was allowed to be anywhere in Rd for some d dimensional function f, then we know how to solve it, we know how to use the gradient descent algorithm. But then now if x is constrained, what can we do? This is the question we are going to think about today.

So, let us start with the following you know a thought experiment, let us say I am trying to solve this problem and then I claim that I have solved this problem, so what I am saying is that I give you a point x star, which is a d dimensional point and then I claim that x star solves this problem. So, I give this point x star to you.

Now, how can you be sure that I am not lying? So, I might be lying, so I am just claiming that x stars solve this problem. Now, given an x star that is claimed to solve let us call this problem some problem A, that is x star is optimal, if I claim that x star is optimal, now how do you check how to check if it is indeed optimal?

In other words, what are the necessary conditions that x star should satisfy, for x star to be optimal? Such the which can be checked easily, so I mean of course the necessary condition is that g(x) star should be less than or equal to 0 and then for any other x, f(x) should be greater than g(x), so any other x that satisfies g(x) is less than or equal to 0, f(x) should be greater than g(x), that is the restating A. So, but that is not easy to check.

Now, is there an easy way to at least check necessity of x star to be an optimal? So, what would you check? So, what is the first thing that you would check to you know if you want to disprove me, what is the first thing that you would want to check? So, pause and think about this. The first thing that you should check is that g(x) star, whichever x star that I give you, is indeed less than or equal to 0.

If g(x) star is greater than 0, then of course it cannot be the optimal for A because the optimal for a necessary should satisfy g(x) is less than or equal to 0 and so we can simply dismiss my claim as bogus. But let us say I am a bit smart, I would give you an x star, which does satisfy g(x) star is less than or equal 0, I know that, that is the first thing you are going to check.

Now, let us say you did check x star g, the functions, the constraints value at x star and you found that g(x) star is less than or equal to 0. Now, how would you what would you do next? So, what is the next thing that you would think about to at least try to if you want to dismiss my claim that x star is the optimal? Let us see what you can do.

Now, if x star is optimal, then I claim that I am going to introduce two terminologies which is not too difficult, but then we will just discuss what it is after they write them down. So, no descent direction should be feasible. So, I claim that this is the next thing that I would check, now but what does this mean?

Well there are two terms here descent direction and feasible direction, so what intuitively what is a descent direction? A descent direction is any direction that reduces the function value, I am at a

current point x star, if I move in some direction and if the functions value reduces then that direction is what I will call a set descent direction.

Now, what is a feasible direction? Well, any direction that takes to a point that is feasible for some steps size, well where are the feasible directions? So, if I am at the current point x star, if I move in some direction, and if I still satisfy the constraint by is feasible I mean that satisfying the constraint in this case it is g(x) is less than or equal to 0, then that direction is a feasible direction I can potentially move in that direction without violating the constraint that is what it means.

So, I can move in that direction, it is still feasible, and now I am saying that the second point is that no descent direction should be a feasible direction. Why is this a good thing to check? Now, let us see what happens if this statement does not hold. If the statement does not hold, then it means that there is some descent direction which is also feasible.

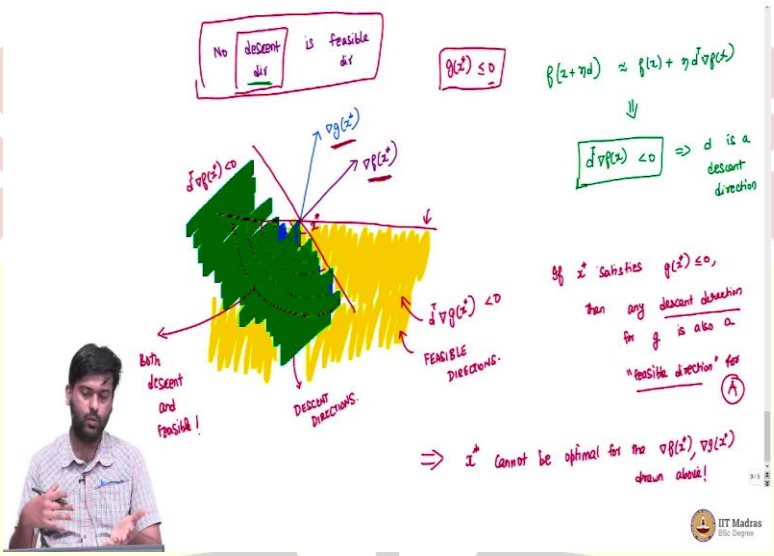
So, that is the negative of this negation of the statement. So, if there is some descent direction which is also feasible, then what does it mean? It means that there is a direction which is both descent and feasible. Well, if there is a direction, which is both descent and feasible, then why is x star not the optimal for this problem? Simple, because if there is a direction which is both descent and feasible, then I can in fact move in that direction.

Now, for a small enough step size, what am I guaranteed, well because it is a descent direction, my function value will reduce. And because it is a feasible direction, I will still satisfy my constraints, which means that I have now moved to a new point where I am still satisfying the constraint and additionally the function value has further reduced which means that the original point that I had which is x star could not be optimal, because the function value at x star is higher than this new point which I could potentially reach if there was a descent direction which is also feasible.

Which means if I claim x star is optimal, then it is necessarily or it better be the case that no descent direction should be a feasible direction. So, this is the intuition that we will want to formalize. So, how would we formalize this idea? So, now remember descent direction is a property of the function f. So, when I say property depends on the function f, so there is nothing g does not play a role at all in determining if a direction is decent or not, because we only care about directions which reduce the functions value.

Whereas, a feasible direction is a property of the function g because the feasibility is only determined by g(x) is less than or equal to 0 the function f does not come into the picture at all. Now, the question is, how do I formulate no descent direction should be a feasible direction mathematically. So, this is what we are going to do next.

(Refer Slide Time: 9:27)



Let me write this again for so that we are clear, no dissent is feasible. So, now let us again go back to pictures. Let us, say you are at some point which I am claiming to be the optimal, let us say it is x star. Now, I want to check if there is a descent direction which is also feasible if there is one, then this cannot be an optimal point. So, necessarily should be the case that no descent direction is feasible.

Now, how would I check that? So, let us first talk about the descent directions. So, where are the descent directions? Well, what does our Taylor series tell us? Well, our Taylor series if you remember tells us f(x +ηd) is approximated as f(x)+ ηd^Tdel f(x) and from this we said that d transpose del f(x)less than 0 implies d is a descent direction.

So, if d satisfies d transpose del f(x) is less than 0 then I can move along d and reduce the function f. So, for the moment I am forgetting the constraint and just looking at the function and seeing where are the descent directions, and it necessarily has to be the case that if d transpose del f(x)is less than 0, then that is a descent direction, which means then if I want to claim x star is optimal, then I should be looking at, are there descent directions of x star?

Of course, there it is going to be descent directions of x star in particular if the gradient del f(x)star points in a certain direction let us say in this direction, then where are the descent directions? Well, the descent directions we just saw earlier that the descent directions are going to be here, any direction which makes a negative dot product with a gradient vector is a descent direction.

So, this is where d transpose del f(x)star is less than 0, in other words these are the descent directions. So, these are the descent directions maybe I will make the x star explicitly clear. So, we have looked at one part of the problem we know where the descent directions are because that is a function of f. Now, we need to understand where the feasible directions.

Now, remember g(x) star is less than or equal to 0, it satisfies the constraint because we checked this already, so if it does not satisfy this constraint, then we can immediately say that x star cannot be optimal, let us say g(x) star satisfies the constraint. Now, we are asking where are the feasible directions. Well, the feasible directions necessarily at least the feasible directions are definitely are going to be those in which if I move if g(x) reduces further it is already less than 0 less than or equal to 0, if it reduces further it can only still be less than or equal to 0.

So, which means that any direction where which is descent for g is also feasible direction. So, let me write that if x star is if x star if the point that I gave you satisfies g(x) star less than or equal to 0, so then any descent direction for g is also a feasible direction, feasible direction for what? Feasible direction for the problem. So, let us call the original problem without we call it A, which is simply minimize f(x)such that g(x) is less than or equal to 0.

So, which means that the feasible directions become descent direction for a g, but we know how to find out descent direction for any function, so how would you do that? We will look at the derivative of that function and then see where are the directions which make a negative dot product. So, let us look at that, so for the point x star, now for the point x star, let us say the derivative was of g remember the same point x star you can take the derivative with respect to f or g.

We saw it with respect to f, when I say derivative, I mean the gradient here, so we the gradient with respect to f was in some direction. Let us, say the gradient with respect to g was pointing in some other direction, so let us say this is del g(x) star. Now, the question is, where are the directions in which you can move and reduce g further?

Well, of course we know where they are, so they are going to be somewhere here, perpendicular to this and if I look at, this directions anything to the below anything below the red line this red line has the property that d transpose then g(x) star is less than 0, in other words, these are the feasible directions for sure. So, now the question is, x star optimal or not?

If my del f(x)star looks in this direction and del g(x) star points in this direction can I simply look at this picture and argue if x star is optimal or not? Now, what does this picture tell me? Well, this picture tells me that I mean what do I want to look at look for what am I looking for I am looking for a direction if I can find a single direction which is a descent direction which is also feasible then this x star cannot be optimal that we already argued.

So, now is there a direction like that in this picture? Well, there clearly is such a direction and where are where is that direction? Well, any direction which is in the intersection of the green and the yellow half spaces is a direction which is both descent both green which is descent and both yellow which is feasible. So, which means that in this case any direction which is here in this sort of say cone let me see if I can color the differently probably not, that is okay.

So, any direction here is both descent and feasible. What is the implication of this? So, this implies if my del f(x)star and del g(x) star of pointing in these two directions like how I have drawn here, which means that for this particular x star, I was my claim was bogus. So, x star cannot be optimal for the for the del f(x)star and del g(x) star drawn for above, for this configuration of del f(x)star and del g(x) star x star cannot be optimal.

Now, then let us ask well, what is the condition which will make this which will make sure that I cannot find a descent direction which is also feasible. In other words, I am asking the question, how should del f(x)star and del g(x) star be related to each other such that I will not be able to find a descent direction which is also feasible? That is the next question we will ask.